



## **TX2 – DATA EXTRACTION AND TRANSFORMATION**

Version 1.4.1

Effective Date: 31 Oct 2025

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# **Invoice Data Submission Specifications**

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**For GST InvoiceNow Requirement**

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**CHANGE LOG**

| Version | Effective Date | Summary of Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0     | 15 April 2024  | - First Release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1.1     | 15 May 2024    | - Annex A (Data Elements) – Revised mandatory elements and simplification of table.<br>- Added a new InvoiceNow-Ready Solution Providers (IRSP) definition and replace PRSP with IRSP.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.2     | 08 Aug 2024    | - Removed Unique Identifier assignment to AP or IRSP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.3     | 04 Dec 2024    | - Provided more clarity to para 2.3 Source of Data and 2.4 List of GST Category Codes Accepted by IRAS<br>- Added new GST Category Code for Purchases Accepted by IRAS “TXNA” in para 2.4.1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.4     | 07 Mar 2025    | - Aligned terminologies: “In-house enterprise solutions”, “Off-the-shelf accounting or finance solutions”, “Invoice data submission”, “Solution-extracted”, “GST InvoiceNow submission”<br>- Replaced “B2C” with aggregated point-of-sale (“POS”) or Simplified Tax Invoice (“STI”) transactions, where applicable<br>- Renamed data flow types to align with IRAS e-Tax Guide.<br>- Moved GST Category Code “NG” and “NR” into a separate table for clarity in para 2.4.1<br>- Updated para 2.5 List of GST Category Codes under the GST InvoiceNow Requirement<br>- Updated Section 4 - Validation Check on Wrongful GST Charges |
| 1.4.1   | 31 Oct 2025    | - Updated the resource hyperlink of the Schematron files in para 2.6.3<br>- Editorial edits in para 2.2.1 and 5.2<br>- Added clarification for ‘Activation’ and ‘Deactivation’ logging in para 7.8                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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## 1. INTRODUCTION

1.1 This document serves as a guide on how the Invoice System, where the invoice data resides, may extract and submit invoice data to their chosen Access Points (“APs”) for onward transmission to IRAS. There are generally two type of entities that have these solutions:

- a) GST-registered businesses who use in-house enterprise solutions (referred as **“Transmitting Enterprises”**)
- b) GST-registered business who use off-the-shelf accounting or finance solutions that are packaged by solution providers, termed as **InvoiceNow-Ready Solution Providers** (referred as **“IRSPs”**). IRSPs are providers with solutions that are pre-integrated to the InvoiceNow network via an IMDA accredited access point provider to support invoice data submission to IRAS. IRSPs are either principal solution providers or resellers with productised add-on modules with direct control over product source codes and intellectual property. The solutions may be:
  - Fully self-developed/wrapped around a technology base code/white-labelled
  - Multi-tenanted/single-tenanted
  - Desktop-based/cloud-based

1.2 The Transmitting Enterprises or the IRSPs are expected to implement a data extraction, transformation and transmission capability to support the invoice data submission. This capability models after how data is extracted for invoicing activity in InvoiceNow. It also means that if these parties are already users of InvoiceNow, the effort to implement requirements to support invoice data submission to IRAS is incremental in nature. Otherwise, the parties will need to engage an IMDA accredited Access Point who will be familiar how data transformation and transmission can be implemented.

Most part of this document will be relevant to both Transmitting Enterprises and IRSPs, unless explicitly indicated.

1.3 This document focuses on technical guidance on enhancements relating to the Invoice system that are the source of invoice data. For specific implementation details on connections to Access Points, the parties should seek further documentation from the Access Points that they are connected to.

The reader may also refer to the TX1 Design Document for a broader understanding of the overall design.

1.4 For information on the implementation of the GST InvoiceNow Requirement, the reader can refer to the [IRAS e-Tax Guide](#) on “Adopting GST InvoiceNow Requirement for GST-registered Businesses”.

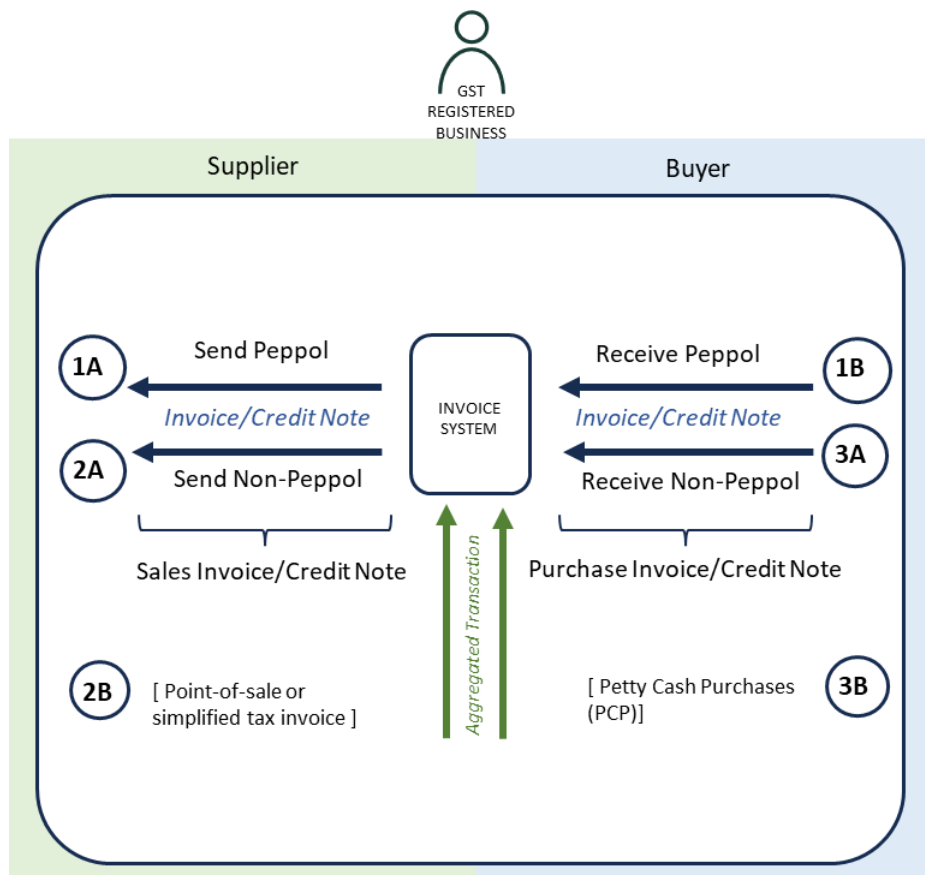
## 2. DATA EXTRACTION

### 2.1 Invoice Data Submission Requirement

2.1.1 IRAS' required invoice/credit note activities to be reported can be defined broadly by 6 data flows representing different directions, transaction channels and point of views, as illustrated in the diagram below.

2.1.2 To align with the data flow types in IRAS' e-Tax guide "Adopting GST InvoiceNow Requirement for GST-registered Businesses", this version shall henceforth follow the naming of their data flows. The mapping between the current data flow in this documentation and IRAS e-Tax guide are summarised as below:

| Type of Submission in IRAS' e-Tax Guide | Corresponding Data Flow in TX2 |
|-----------------------------------------|--------------------------------|
| Type 1A Submission                      | Type 1A (f.k.a. Type A)        |
| Type 1B Submission                      | Type 1B (f.k.a. Type D)        |
| Type 2 Submission                       | Type 2A (f.k.a. Type B)        |
|                                         | Type 2B (f.k.a. Type C)        |
| Type 3 Submission                       | Type 3A (f.k.a. Type E)        |
|                                         | Type 3B (f.k.a. Type F)        |



## DATA EXTRACTION AND TRANSFORMATION

2.1.3 Flow Type 1A, 2A and 2B are part of the Supplier process where businesses issue invoices/credit notes to their customers (Type 1A and 2A) and where aggregated point-of-sale (“POS”) or simplified tax invoice (“STI”) transactions are recorded in their systems (Type 2B).

2.1.4 Correspondingly, flow type 1B, 3A and 3B are part of the Buyer process where businesses receive invoices/credit notes from their supplier (Type 1B and 3A) and where aggregated data from petty cash purchases are recorded in their systems (Type 3B).

## 2.2 Data Packages

2.2.1 The Transmitting Enterprises or the IRSPs are to submit data packages representing each of these data flows, as described in the following table.

| Data Flow | Peppol/ Non-Peppol | Item                                                                                                        | Type of Transaction | Granularity of data to be sent                                                                                                                                                                                                                           | When invoice data is sent to IRAS system                                                                 | Specifications                                            |
|-----------|--------------------|-------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Type 1A   | Peppol             | Issued Invoices /Credit Note                                                                                | Supply              | Transaction level (Single)                                                                                                                                                                                                                               | When invoice is issued                                                                                   | Peppol Transaction (PINT@SG)                              |
| Type 2A   | Non-Peppol         | Issued Invoices /Credit Notes                                                                               | Supply              | Transaction level (Single)                                                                                                                                                                                                                               | The transactions should be submitted to IRAS system before or at the same time when the return is filed. | Solution-Extracted (Data specifications based on PINT@SG) |
| Type 2B   | Non-Peppol         | Aggregated point-of-sale (“POS” <sup>1</sup> ) or simplified tax invoice (“STI” <sup>2</sup> ) transactions | Supply              | Aggregated level<br>Such as daily cash register sales or sales made via point-of-sale systems (tagged as “POS”), or simplified tax invoices issued (tagged as “STI”) consolidated in a single invoice and submitted to IRAS system on a regular interval |                                                                                                          |                                                           |
| Type 1B   | Peppol             | Accepted Invoices/Credit Notes                                                                              | Purchase            | Transaction level (Single/Bulk)                                                                                                                                                                                                                          |                                                                                                          |                                                           |
| Type 3A   | Non-Peppol         | Accepted Invoices/Credit Notes                                                                              | Purchase            | Transaction level (Single/Bulk)                                                                                                                                                                                                                          |                                                                                                          |                                                           |

<sup>1</sup> Supplies made via point-of-sale (“POS”) systems. Cash registers are also considered as POS systems.

<sup>2</sup> Supplies where simplified tax invoices (“STI”) are issued, even if they are not generated from POS systems. A simplified tax invoice can be issued instead of a regular tax invoice if the total amount payable for your supply (including GST) does not exceed \$1,000.

## DATA EXTRACTION AND TRANSFORMATION

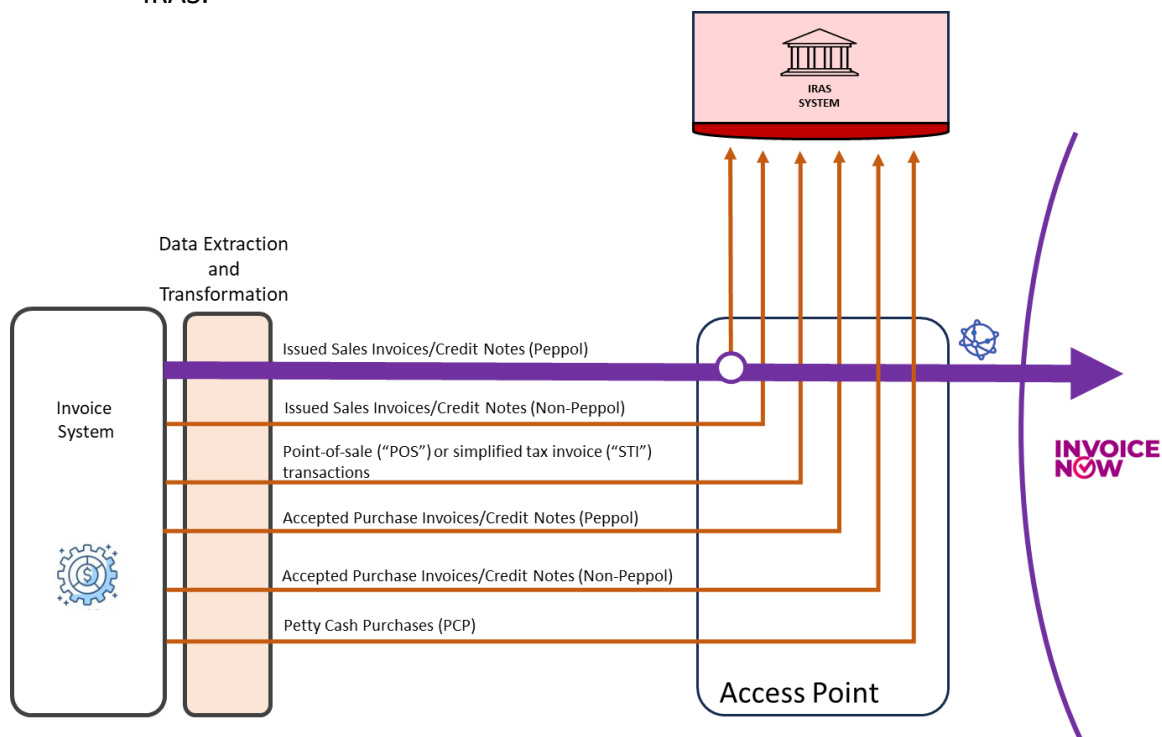
| Data Flow | Peppol/<br>Non-Peppol | Item                                                         | Type of<br>Transact-<br>ion | Granularity of data<br>to be sent                                                                                                                          | When invoice data<br>is sent to IRAS<br>system | Specifications |
|-----------|-----------------------|--------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|
| Type 3B   | Non-Peppol            | Aggregated<br>Petty Cash<br>Purchases<br>(PCP <sup>3</sup> ) | Purchase                    | Aggregated level<br>Such as petty cash<br>purchases<br>consolidated in a<br>single invoice that is<br>submitted to IRAS<br>system on a regular<br>interval |                                                |                |

2.2.2 The table describes two types of data structures depending on how data can be obtained from the source:

- Peppol Transaction (Type 1A) – refers to near real time transmission of invoices over InvoiceNow where copies of them will be transmitted to IRAS
- Solution-Extracted (Type 1B, 2A, 2B, 3A and 3B) – refers to invoice data extracted from Invoice System and packaged for transmission to IRAS

For “Peppol” Transaction data, the complete Peppol document, including the SBDH, will be submitted to IRAS as is with no modification. For “Solution Extracted” data, refer to detailed specification on data structure in Annex A and B.

2.2.3 Once relevant invoice data are extracted and transformed according to the specifications of the data packages, they can be passed to Access Point to transmit to IRAS.



<sup>3</sup> Petty cash purchases (“PCP”). Staff claims and purchases using corporate cards are also considered as “PCP” transactions.

## 2.3 Source of Data

- 2.3.1 While some IRSPs or Transmitting Enterprises with a central Invoice System may have all the data required for both inbound and outbound residing in one or more data tables within the same system, there may be others which may be using separate systems to manage inbound and outbound invoices. In addition, businesses may also be managing different types of goods and services transactions through several outbound billing systems. These businesses will have to consider the most optimal approach to extract the required information. For example, businesses with multiple systems can consider funnelling data into a central system or staging area/middleware before performing the transmission to IRAS. The system design should also take into consideration businesses' need to connect through multiple Access Points, if necessary.

### Transmission of Invoice Data

- 2.3.2 Under the GST InvoiceNow Requirement, the invoice data that businesses need to transmit to IRAS includes data pertaining to both invoices and credit notes. Transactions to be transmitted to IRAS under the requirement shall follow the relevant guidelines provided to perform the submission.

## 2.4 List of GST Category Codes Accepted by IRAS

IRAS has established a comprehensive list of prescribed GST category codes for businesses to classify their supplies and purchase transactions. IRSPs or Transmitting Enterprises should incorporate the standardised set of GST Category Codes into their InvoiceNow-Ready Solutions or systems and use it to tag their transactions for submission to IRAS as **only this set of category codes will be accepted by IRAS**. Please refer to the tables below for more details.

### GST Category Codes for Supplies Accepted by IRAS<sup>4</sup>

| Types of supplies | GST category code | GST rate | Description                                 |
|-------------------|-------------------|----------|---------------------------------------------|
|                   | SR                | 9%       | Standard-rated supply of goods or services  |
|                   | SRCA-S            | NA       | Customer accounting supply made by supplier |

<sup>4</sup> The comprehensive list of "GST Category Codes Accepted by IRAS" can be found in Annex E of IRAS e-Tax Guide.



## DATA EXTRACTION AND TRANSFORMATION

| Types of supplies       | GST category code | GST rate | Description                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard-rated supplies | SRCA-C            | 9%       | Customer accounting supply accountable by the customer on supplier's behalf                                                                                                                                                                                                                         |
|                         | SRLVG             | 9%       | Own supply of Low-Value Goods ("LVG")                                                                                                                                                                                                                                                               |
|                         | NA                | NA       | Taxable supplies where GST need not be charged under specific GST schemes such as Gross Margin Scheme ("GMS"), Approved Marine Fuel Trader ("AMFT") Scheme, Approved 3rd Party Logistics ("A3PL") Scheme. Please refer to the respective e-Tax Guides for invoicing and GST reporting requirements. |
|                         | SRRC              | 9%       | Imported services and LVG accountable by the GST-registered customer under reverse charge                                                                                                                                                                                                           |
|                         | SROVR-RS          | 9%       | Supply of remote services accountable by the electronic marketplace on behalf of third-party suppliers                                                                                                                                                                                              |
|                         | SROVR-LVG         | 9%       | Supply of LVG accountable by the redeliverer or electronic marketplace on behalf of third-party suppliers                                                                                                                                                                                           |
|                         | DS                | 9%       | Deemed supplies                                                                                                                                                                                                                                                                                     |
| Zero-rated supplies     | ZR                | 0%       | Zero-rated supplies                                                                                                                                                                                                                                                                                 |
| Exempt supplies         | ES33              | NA       | Regulation 33 Exempt Supplies                                                                                                                                                                                                                                                                       |
|                         | ESN33             | NA       | Non-Regulation 33 Exempt Supplies                                                                                                                                                                                                                                                                   |
| Out-of-scope supplies   | OS                | NA       | Supplies outside the scope of the GST Act                                                                                                                                                                                                                                                           |

### Supplies made by non-GST registered business

| Types of supplies     | GST category code | GST rate | Description                                                                                                                                                        |
|-----------------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Out-of-scope supplies | NG                | NA       | Supplies made by non-GST registered business, i.e. the non-GST registered business is not required to determine GST treatment, charge GST or file any GST returns. |

**GST Category Codes for Purchases Accepted by IRAS<sup>5</sup>**

| <b>Types of purchases</b> | <b>GST category code</b> | <b>GST rate</b> | <b>Description</b>                                                                                                                                                                                                                                                          |
|---------------------------|--------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standard-rated purchases  | TX                       | 9%              | Standard-rated taxable purchases                                                                                                                                                                                                                                            |
|                           | IM                       | 9%              | Import of goods (9% GST paid to Singapore Customs on the import of goods into Singapore)                                                                                                                                                                                    |
|                           | ME                       | 0%              | Import of goods under the Major Exporter Scheme (“MES”), A3PL Scheme or other approved schemes                                                                                                                                                                              |
|                           | IGDS                     | 9%              | Import of goods under the Import GST Deferment Scheme (“IGDS”)                                                                                                                                                                                                              |
|                           | TXCA                     | 9%              | Standard-rated purchases of prescribed goods subject to customer accounting                                                                                                                                                                                                 |
|                           | TXNA                     | NA              | Purchases made under specific GST schemes, such as Gross Margin Scheme (“GMS”), Approved Marine Fuel Trader (“AMFT”) Scheme, Approved 3rd Party Logistics (“A3PL”) Scheme.<br><br>Please refer to the respective e-Tax Guides for invoicing and GST reporting requirements. |
|                           | TXRC-TS                  | 9%              | Imported services and LVG claimable by the GST-registered customer under reverse charge                                                                                                                                                                                     |
|                           | TX-ESS                   | 9%              | Standard-rated purchases directly attributable to Regulation 33 exempt supplies                                                                                                                                                                                             |
|                           | TXRC-ESS                 | 9%              | Imported services and LVG claimable by the GST-registered customer under reverse charge that are directly attributable to Regulation 33 exempt supplies                                                                                                                     |
|                           | IM-ESS                   | 9%              | Import of goods with GST paid to Singapore Customs that are directly attributable to Regulation 33 exempt supplies                                                                                                                                                          |
|                           | TX-N33                   | 9%              | Standard-rated purchases directly attributable to non-Regulation 33 exempt supplies                                                                                                                                                                                         |
|                           | TXRC-N33                 | 9%              | Imported services and LVG claimable by the GST-registered customer under reverse charge that are directly attributable to non-Regulation 33 exempt supplies                                                                                                                 |

<sup>5</sup> The comprehensive list of “GST Category Codes Accepted by IRAS” can be found in IRAS e-Tax Guide Annex E.

## DATA EXTRACTION AND TRANSFORMATION

| Types of purchases       | GST category code | GST rate | Description                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | IM-N33            | 9%       | Import of goods with GST paid to Singapore Customs that are directly attributable to non-Regulation 33 exempt supplies                                                                                                                                                                        |
|                          | TX-RE             | 9%       | Residual input tax – purchases from GST-registered suppliers that are subject to GST at 9% and are either: <ul style="list-style-type: none"> <li>- Attributable to the making of both taxable and exempt supplies; or</li> <li>- Incurred for the overall running of the business</li> </ul> |
|                          | TXRC-RE           | 9%       | Imported services and LVG claimable by the GST-registered customer under reverse charge that are residual                                                                                                                                                                                     |
|                          | IM-RE             | 9%       | Import of goods with GST paid to Singapore Customs that are residual                                                                                                                                                                                                                          |
| Zero-rated purchases     | ZP                | 0%       | Zero-rated purchases                                                                                                                                                                                                                                                                          |
| Non-reportable purchases | BL                | 9%       | Disallowed expenses                                                                                                                                                                                                                                                                           |
|                          | EP                | NA       | Exempt purchases                                                                                                                                                                                                                                                                              |
|                          | OP                | NA       | Out-of-scope purchases received from GST-registered suppliers<br><br>Purchases from GST-registered suppliers where input tax is not claimed (e.g. not for business purposes, invalid tax invoices, not claiming input tax out of prudence etc.)                                               |
|                          | NR                | NA       | Purchases received from non-GST registered suppliers                                                                                                                                                                                                                                          |

**Purchases received by non-GST registered business**

| <b>Types of purchases</b> | <b>GST category code</b> | <b>GST rate</b> | <b>Description</b>                                                                                                                                                      |
|---------------------------|--------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-reportable purchases  | NR                       | NA              | Purchases received by non-GST registered business, i.e. the non-GST registered business is not allowed to claim any input tax and not required to file any GST returns. |

- 2.4.1 Businesses that are using their existing internal GST category codes can continue to use their internal tax codes if they prefer, but IRSPs or Transmitting Enterprises would need to assist in mapping their internal tax/category codes to IRAS' GST Category Codes in order to submit invoice data to IRAS.
- 2.4.2 Alternatively, businesses can also consider adopting IRAS' list of prescribed GST category codes to classify their transactions moving forward.

**2.5 List of GST Category Codes under the GST InvoiceNow Requirement**

- 2.5.1 To simplify the data collection rules under the GST InvoiceNow Requirement, GST-registered businesses are required to transmit to IRAS invoice data **reported in the GST return**<sup>6</sup>, other than specific excluded transactions ("Excluded Transactions").

**What Businesses need to transmit under the GST InvoiceNow Requirement:**

- All standard-rated supplies
- All zero-rated supplies
- All exempt supplies
- All standard-rated purchases
- All zero-rated purchases
- Except Excluded Transactions

For more details, please refer to IRAS e-Tax guide.

<sup>6</sup> This replaces the draft scope of transactions in the draft e-tax guide published during the Public Consultation.

2.5.2 Please refer to the tables below for more details.

**Invoice Data Reported in the GST Return – Mandatory to Transmit to IRAS**

| Type of Supplies/<br>Purchases | Examples of Data to Transmit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GST Category Codes                                                                                                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Standard-rated supplies        | <p>Local sales of goods and services:</p> <ul style="list-style-type: none"> <li>Supplies where GST is disregarded or suspended under certain GST schemes e.g., Approved Contract Manufacturer and Trader (“ACMT”) or Approved Marine Fuel Trader Scheme (“AMFT”)</li> <li>Supplies where GST is accounted on a different basis other than at prevailing GST rate on gross sales value, e.g., Gross Margin Scheme (“GMS”), Discounted sales price scheme</li> <li>Supplies where Customer Accounting is required to be applied</li> <li>Supplies of remote services and LVG accountable by the electronic marketplace on behalf of third-party suppliers</li> <li>Own supplies of LVG</li> </ul> | <ul style="list-style-type: none"> <li>SR</li> <li>NA</li> <li>SRCA-S</li> <li>SROVR-RS</li> <li>SROVR-LVG</li> <li>SRLVG</li> </ul> |
| Zero-rated supplies            | <ul style="list-style-type: none"> <li>Export sales of goods</li> <li>Provision of international services</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>ZR</li> </ul>                                                                                 |
| Exempt supplies                | <ul style="list-style-type: none"> <li>Sale or lease of residential properties</li> <li>Local sale of investment precious metals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>ES33</li> <li>ESN33</li> </ul>                                                                |
| Standard-rated purchases       | <ul style="list-style-type: none"> <li>Purchases with input tax claims, e.g. purchase of business inventories from GST-registered suppliers</li> <li>Purchases where input tax is directly attributable to Regulation 33 exempt supplies and non-Regulation 33 exempt supplies</li> <li>Purchases where input tax claims are residual in nature</li> <li>Purchases under certain GST schemes e.g. GMS, AMFT and Approved 3rd Party Logistics Company (“A3PL”) scheme</li> <li>Purchases where Customer Accounting is applied</li> </ul>                                                                                                                                                          | <ul style="list-style-type: none"> <li>TX</li> <li>TX-ESS</li> <li>TX-N33</li> <li>TX-RE</li> <li>TXNA</li> <li>TXCA</li> </ul>      |

| Type of Supplies/<br>Purchases | Examples of Data to Transmit                                                                                                                                                        | GST Category Codes                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Zero-rated purchases           | <ul style="list-style-type: none"> <li>• Purchase of goods that are exported</li> <li>• Purchase of international services e.g. international freight charges, IDD calls</li> </ul> | <ul style="list-style-type: none"> <li>• ZP</li> </ul> |

### **Submission of invoice data with supplies or purchases that fall within and outside the requirement**

- 2.5.3 If the invoice or credit note has a line item that pertains to invoice data that falls within the scope of the GST InvoiceNow Requirement above, then the invoice data relating to the entire invoice or credit note has to be transmitted to IRAS under the requirement.

Specifically for purchases invoice data, only received invoices and credit notes that have been approved in the invoice system or accepted for payment should be sent to IRAS.

### **Option to Aggregate Invoice Data for Transmission**

- 2.5.4 Where it is not feasible to submit invoice data on a transactional basis, businesses can choose to aggregate the following transactions on a regular basis before transmitting to IRAS:
- (a) Supplies made via point-of-sale systems ("POS" transactions). Cash register sales would fall under this category.
  - (b) Supplies where simplified tax invoices are issued, even if they are not generated from POS systems ("STI" transactions).
  - (c) Petty cash purchases ("PCP" transactions). Staff claims and purchases using corporate cards would fall under this category.

Such aggregated transactions can be consolidated into a single line item (tagged with the applicable GST category code above) and submitted in the form of solution-extracted invoice data on a regular basis (e.g. weekly, monthly) as per the Type 2B and Type 3B data flows in the diagram under Para 2.1.2.

### **Option to Transmit All Supplies and Purchases Invoice Data**

- 2.5.5 Businesses may opt to transmit all supplies and purchases invoice data, including those not under the requirement if doing so eases their compliance burden. For instance, businesses may opt to transmit excluded transactions such as deemed supplies, non-sales exports, goods and services under reverse charge. Such transactions will also be accepted by IRAS if the category codes of the transactions are within the "List of GST Category Codes Accepted by IRAS" at Para 2.4.1.

### 2.6 Data schema Validation Check

- 2.6.1 IRSPs and Transmitting Enterprises that are submitting data to IRAS system must ensure that data entered are compliant with the mandatory data elements as required in GST InvoiceNow Requirement.
- 2.6.2 The invoice data for transmission to IRAS shall follow a Peppol like syntax where some optional Peppol elements are made mandatory. IRSPs and Transmitting Enterprises should ensure that all mandatory data elements are provided for transmission to IRAS. These mandatory field details can be found in Annex A and B.
- 2.6.3 The Schematron files can be downloaded [here](#) so that IRSPs and Transmitting Enterprises can use it to validate and make sure invoices submitted to IRAS are in the correct xml structure. IRSPs and Transmitting Enterprises may utilise the Schematron files to validate their invoice data extraction when developing their solution for IRAS submission. Where possible, IRSPs and Transmitting Enterprises should validate any invoice being generated and handle any errors.

### 2.7 Frequency of Submission

- 2.7.1 For Type 1A (Peppol invoice method), submission happens immediately after C2 Access Point successfully delivering the document to C3 Access Point. In most cases, this should be within minutes, but it should not exceed an hour.
- 2.7.2 For other data flow types (Solution-Extracted method), the transactions should be submitted to IRAS system before or at the same time when the return is filed. This means that the transactions should be submitted to IRAS system no later than the return due date (along with the return). However, such document data extraction is recommended to be carried out on a regular basis such as daily or weekly, after the document is booked in the system for issuance to their business partner. This will allow the businesses sufficient time to correct any submission errors by spreading the reconciliation effort over the GST filing period. This also help to avoid peak time loading on systems involved.

### 2.8 Inclusion of attachment

- 2.8.1 For Peppol outbound invoices transmitting to IRAS, attachment should be kept if attachment is included while sending to the customer.
- 2.8.2 For all other invoices that are Solution-Extracted, no attachment is required to be included, only invoice data shall be included in the submission data.

### 2.9 Size limit

- 2.9.1 IRAS API supports a data package up to 10 MB in size. For data flow A, where Peppol Access Points can support to a minimum of 100MB, the auto submission on the invoice data package above 10MB to IRAS system will fail. In such situation, a recovery mechanism should be activated using solution extracted method.

- 2.9.2 Access Point will be required to provide the IRSP or Transmitting Enterprises with the failure code stating reason for the rejection.
- 2.9.3 IRSP or Transmitting Enterprises would need to re-submit the invoice data without attachment to IRAS system.
- 2.9.4 The IRSP or Transmitting Enterprises should ensure that they do not re-send the document via the Peppol network a second time.

### 2.10 Assigning Counterparty Label for Non Peppol Delivered Documents

- 2.10.1 Invoice Systems are required to store the mode of sending and receipt of invoices to be able to differentiate data flow between type 1A and 2A, and type 1B and 3A to meet the submission requirement.
- 2.10.2 To support this requirement, a special label is introduced to denote the supplier or customer involved in a Non Peppol delivery invoice/credit note. The label is derived as follows:

Format: 0195:C5UIDYYYYYYYYYY

Where:

‘YYYYYYYYYY’ is the UEN of the customer (if known) or else can be the first five characters of the name (without spaces) plus a random five digit alphanumeric.

| Data Flow | Supplier             | Customer             |
|-----------|----------------------|----------------------|
| Type 1A   | Peppol ID            | Peppol ID            |
| Type 2A   | Peppol ID            | 0195:C5UIDYYYYYYYYYY |
| Type 1B   | Peppol ID            | Peppol ID            |
| Type 3A   | 0195:C5UIDYYYYYYYYYY | Peppol ID            |

## 3. DATA TRANSFORMATION AND TRANSMISSION

### 3.1 Data Transformation

- 3.1.1 The data transformation involves packaging the data into an XML format that complies to IRAS’ requirement as indicated in Section 2. The specifications can be found in Annex A and B.
- 3.1.2 The complete packaging of the data for submission includes preparing the envelop information, called the SBDH (Standard Business Document Header).
- 3.1.3 In some situation, the Access Point providers may offer data transformation services based on agreed data exchange format between the party and the Access Point (for example, data in JSON or CSV format). There may also be situations where the party



prepares the XML while the Access Point complete the work with the SBDH. For detailed arrangement, please discuss with your Access Point provider. Similar to how an InvoiceNow compliant invoice is generated, a service provider may create the files on behalf of a business. However, the responsibility to provide all the required data elements remains with the GST-registered business.

### 3.2 Data Transmission

- 3.2.1 The data transmission between IRSP or GST-registered business and Access Point is typically done in two popular ways; an API call or SFTP. Firewall and network configuration may be required to allow connection to the Access Point. For detailed arrangement, please discuss with your Access Point provider.

## 4. VALIDATION CHECK ON WRONGFUL GST CHARGES

- 4.1 As part of the implementation for the GST InvoiceNow Requirement, IRAS is enabling IRSPs and Transmitting Enterprises to build the validation check on wrongful GST charges feature in their InvoiceNow-Ready Solution.
- 4.2 This check utilises invoice data such as the Supplier GSTN, Invoice Date and GST Amount and data returned from the IRAS' Check GST Register API to detect invoices that wrongly charge GST.
- 4.3 Implementing this validation check in your InvoiceNow-Ready Solution will help make tax compliance easier for you or your clients; it will also provide you with the assurance that the GST charges on issued invoices are legitimate.
- 4.4 IRSPs and Transmitting Enterprises may also wish to check with their Access Points as some Access Points may provide a passthrough API service to enable them to consume IRAS' Check GST Register API and build the validation check.
- 4.5 For more information on the specifications of this recommended feature, please refer to [Recommended Features for the GST InvoiceNow Requirement](#).

## 5. TRACEABILITY AND TRACKING OF SUBMISSIONS

- 5.1 SBDH is required for all documents submitted to IRAS (both Peppol and Solution Extracted documents). To ensure that every document and every transmission of content on the network is being uniquely identified for tracking purposes we have two UUIDs:
- i. **Document UUID** - which identifies the invoice/credit note document, and
  - ii. **Transmission UUID** - which identifies the unique transmission of the document over the network.

UUID should be represented in hexadecimal value, eg.

bfeac3d0-82d8-4195-9d88-53b3547f83a3

### 5.2 Document UUID

For sales invoices, a UUID is required to be assigned as a system field and should be included in the Peppol Invoice sent to the buyer as well as IRAS.

The field to store this UUID is [ubl:Invoice/cbc:UUID](#) or [ubl:CreditNote/cbc:UUID](#).

For purchasing invoices received via Peppol network, the Invoice System should record the original UUID value, and when the purchasing invoices are extracted for submission to IRAS, the UUID value from the original supplier invoice should be included.

For purchasing invoices received outside Peppol Network, recording the UUID value from the supplier invoice is recommended if feasible.

### 5.3 Embedding Document UUID into SBDH

In order for AP to be able to maintain traceability, the document UUID should also be appended to SBDH in the following sections as illustrated below:

```
<BusinessScope>
  <Scope>
    <Type>DOCUMENT UUID</Type>
    <InstanceIdentifier>bfeac3d0-82d8-4195-9d88-53b3547f83a3</InstanceIdentifier>
  </Scope>
</BusinessScope>
```

### 5.4 Transmission UUID

A UUID is expected to be populated in the DocumentIdentification/InstanceIdentifier field to identify the transmission of documents on the network. The UUID MUST be unique for each Business Message Envelope being created.

The UUID MUST be globally unique e.g. 118e3040-51d2-11e3-8f96-0800200c9a66

The UUID should be generated by the party packaging the invoice data to be encoded into the SBDH of the Solution-Extracted invoice data.

```
<DocumentIdentification>
  <Standard>urn:oasis:names:specification:ubl:schema:xsd:Invoice-2</Standard>
  <TypeVersion>2.1</TypeVersion>
  <InstanceIdentifier>553a0065-b2c0-40c6-bf63-fd1463d59478</InstanceIdentifier>
  <!-- reduced instance file>
</DocumentIdentification>
```

### 5.5 Packaging the SBDH

|                   | Peppol Delivered                                                                                                | Solution Extract  |                                                                                 |                    |                   |                            |
|-------------------|-----------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------|--------------------|-------------------|----------------------------|
|                   | Peppol Inv/CN                                                                                                   | Non-Peppol Inv/CN | Aggregated point-of-sale (“POS”) or simplified tax invoice (“STI”) transactions | Peppol Inv/CN      | Non-Peppol Inv/CN | Petty Cash Purchases (PCP) |
| Sender            | Supplier Peppol ID                                                                                              |                   |                                                                                 | Customer Peppol ID |                   |                            |
| Receiver          | Customer Peppol ID                                                                                              | “IRAS”            |                                                                                 |                    |                   |                            |
| Transmission UUID | Unique identifier of the SBDH/transmission, located at <code>DocumentIdentification / InstanceIdentifier</code> |                   |                                                                                 |                    |                   |                            |
| Document_UUID     | NA                                                                                                              | UUID of payload   |                                                                                 |                    |                   |                            |

The above table defines the content to be included in the SBDH according to the data packages.

- Sender - The sender on the SBDH denotes the GST-Registered Business submitting (aka "Submitter") the invoice data to IRAS system. Use Peppol IDs of the Submitter.
- Receiver - For Peppol Standard package, the original Peppol document is sent as is, where Receiver will be the Customer's Peppol ID. For all the Solution Extracted packages, the Receiver will be "IRAS".

Access Point providers will reject any invoice data to IRAS if the SBDH of the package is incorrectly constructed.

### 5.6 IRAS's Acknowledgment ID

IRAS's API will return a unique identifier after every successful invoice submission. The Invoice System needs to store the acknowledgement ID and status of each transmission as part of the system logging.

## 6. REPORTING

- 6.1 IRSPs and Transmitting Enterprises should provide logging and reconciliation facilities to keep track of transmission of data to IRAS. These logs should be stored for a period in accordance with IRAS' record keeping requirement. This will also enable users to troubleshoot and reconcile any mismatches or issues with GST claims submissions.
- 6.2 IRSPs and Transmitting Enterprises should use reports from Access Points to reconcile and validate GST claims submissions. Reports from Access Point are expected to contain the following:
- Date and time of invoice data submission
  - Unique message Instance Identifier (UUID) of the transmission
  - Unique message Instance Identifier (UUID) of the document
  - Peppol ID of the entity submitting the invoice data
  - Status of the submission of the invoice data to IRAS
- 6.3 Periodically, IRSPs or Transmitting Enterprises may be asked by IRAS to provide evidence of invoice data submission using a UUID as the unique reference.
- 6.4 The data to aid tracing of submissions shall be kept according to IRAS' requirement to keep proper records and accounts. Please refer to link for details - <https://www.iras.gov.sg/taxes/goods-services-tax-gst/basics-of-gst/invoicing-price-display-and-record-keeping/keeping-records>
- 6.5 Note that the Access Point will not hold in storage invoice/credit note data (unless with their customer's explicit consent). The Access Points may keep records of all the UUIDs they have sent to IRAS for reporting and tracing. The Access Point may avail such reports to the GST-registered businesses.

## 7. ONBOARDING AND ACTIVATION

- 7.1 There are 3 stages for any GST-registered businesses to submit data to IRAS system.
- a. Register on InvoiceNow network
  - b. Activating invoice data submission
  - c. Start transmitting invoice data
- 7.2 As a pre-condition to submit the invoice data to IRAS system through the InvoiceNow accredited Access Points, all GST-registered businesses will need to be registered on the Peppol network.
- All IRSPs should have InvoiceNow registration capability. Transmitting Enterprise customers can leverage on their chosen Access Point for the same.
- 7.3 To activate for invoice data submission to IRAS, the end user is required to complete additional authorization step using CorpPass. The activation feature is available by your Access Points provider as a pass-through service to trigger the activation process.

After completing the activation process, GST-registered business can start submitting invoice data to IRAS.

- 7.4 For Transmitting Enterprises, they will also work with their Access Points provider to activate invoice data submission. Once the activation is completed, the Access Point will allow the submission of invoice data to IRAS system.
- 7.5 IRSPs are required to provide a function for their GST-registered business customers to activate GST InvoiceNow submission feature. This function will tap on the passthrough API that the Access Point will provide. Upon successful completion of invoice data submission activation, IRSP shall maintain the activation status in their solution as an indicator to decide if their customer's invoice/credit note data should be submitted to IRAS system.
- 7.6 Like activation, there will also be the requirement to support de-activation. A corresponding passthrough API will be provided by the Access Point. Likewise, the IRSPs are required to use deactivation indicators to block sending of invoice data to IRAS' system.
- 7.7 IRSP should adopt the term 'GST InvoiceNow submission' in their user interfaces for their invoice data submission feature such as activation, submission, reporting etc.
- 7.8 IRSPs are required to implement and maintain logs for the activation and deactivation of the GST InvoiceNow by recording the timestamp, login user ID, action type, status, and error information.